

Covid-19 Acoustic Detector

Milestone 1

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— Requirements



Functional

- Guided three-cough recording via microphone
- Cough-presence gate, then segmentation
- CNN inference with Low / Medium / High risk + confidence
- Disclaimers and recommendations on every result
- Local history and trends dashboard (export CSV/JSON)

Non Functional

- Capture to result under 6s; inference under 1.5s
- Stateless server: no health data retained
- handling of bad input
- Accessible UI (color + text), modern browsers
- Ethics: pre-screening signal, never a diagnosis

Test Plan (functional)

Guided three cough recording

- Clear mic permission prompt + live level meter
- Always captures exactly 3 coughs, auto-stops after 3rd

Cough-presence gate, then segmentation

- Rejects non cough audio (speech, noise) before analysis runs
- Slices each cough into a precise 1 second clip

CNN inference with risk + confidence

- Same input always gives consistent result (no randomness)
- Below 0.8 confidence → shows "inconclusive," not a guess

Disclaimers on every result

- "Not a diagnosis" notice shown on every screen, no exceptions
- Recommendation text always matches the risk level

Test Plan (non functional)

Speed (under 6s / 1.5s)

- Full flow timed across 20 real runs
- AI analysis alone must stay under 1.5s

Stateless server

- Server storage checked after each run, confirms nothing is kept

Handling bad input

- Corrupted, silent, or oversized files must fail gracefully, never crash

Accessibility + ethics

- Risk shown with color *and* text, works across major browsers
- Wording never implies a medical diagnosis

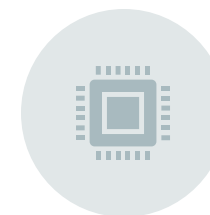
Milestone 1 Matrix

Task	Done	Drew	Amanda	Loleyi	Richard	To do
Requirements & project plan	100%	10%	70%	10%	10%	N/A
Literature review	100%	10%	70%	10%	10%	N/A
Data preprocessing & MFCC/Mel	100%	70%	10%	10%	10%	N/A
Audio augmentation	100%	70%	10%	10%	10%	N/A
Baseline PyTorch CNN	100%	10%	10%	70%	10%	N/A
React recording interface	100%	10%	10%	10%	70%	N/A

Resolve Technical Problems



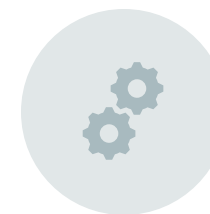
Data Preprocessing & Augmentation - Standardizing inconsistent cough recordings while creating realistic augmented samples for better generalization.



CNN Development & Overfitting - Selecting and tuning the CNN architecture while preventing strong training performance from translating into poor results on unseen data.



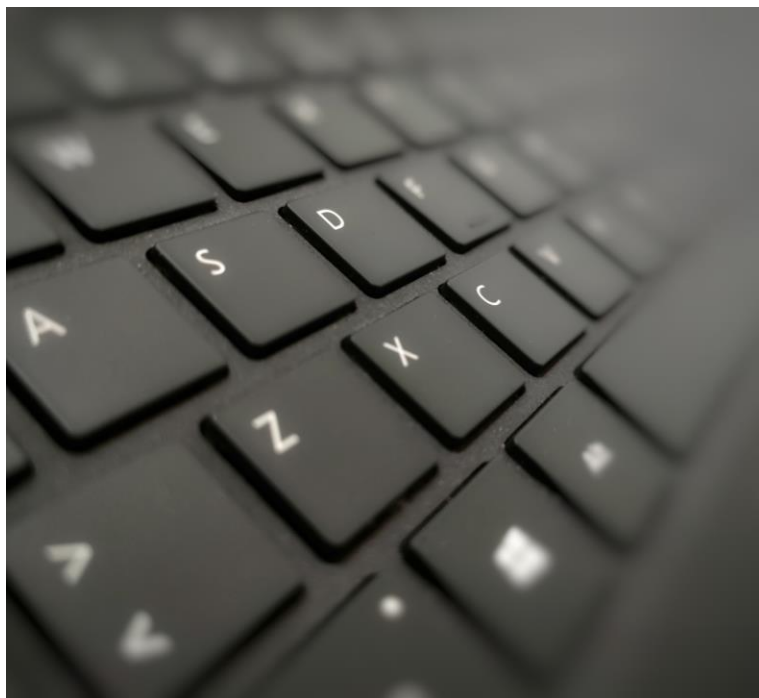
Cough Segmentation & XGBoost Integration - Reliably identifying cough events and connecting the cough-presence gate to the CNN pipeline.



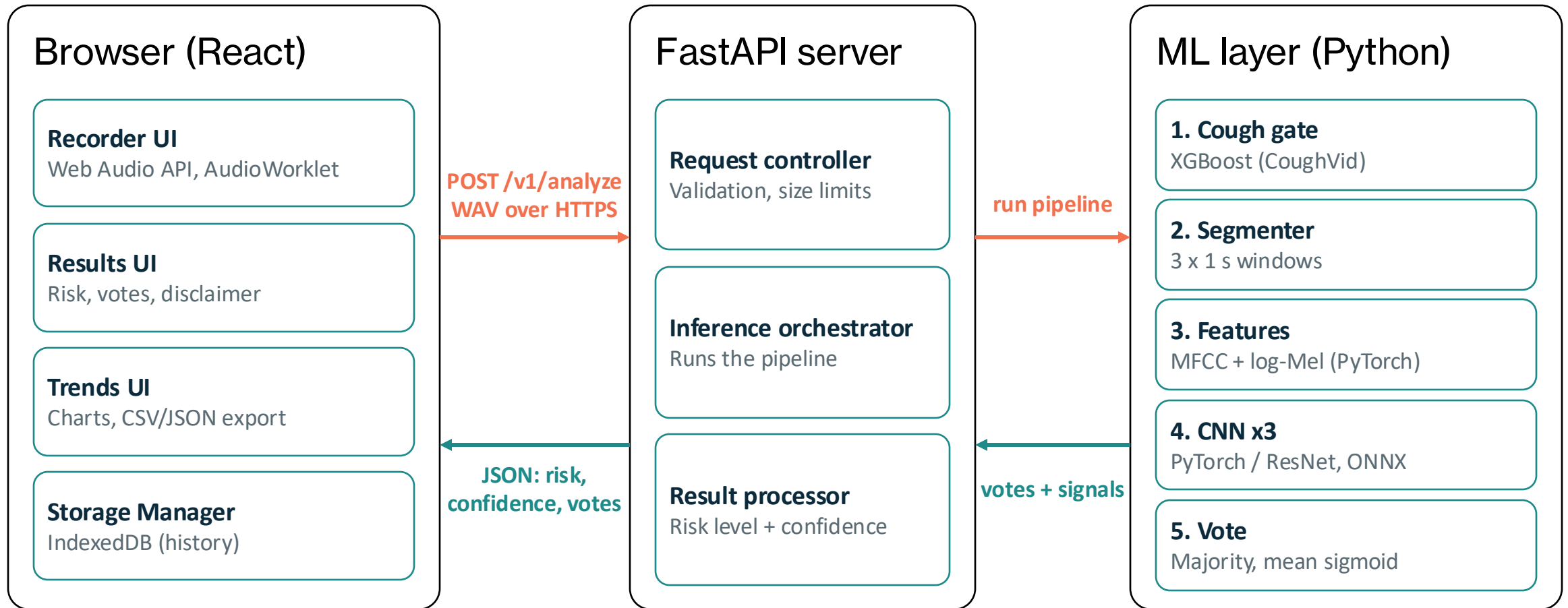
React/FastAPI Integration & Performance - Connecting browser audio capture to the backend inference pipeline while keeping the complete analysis fast and reliable.



Class Imbalance - The dataset does not contain an equal number of examples for each target class, which can cause the CNN to favor more represented classes.

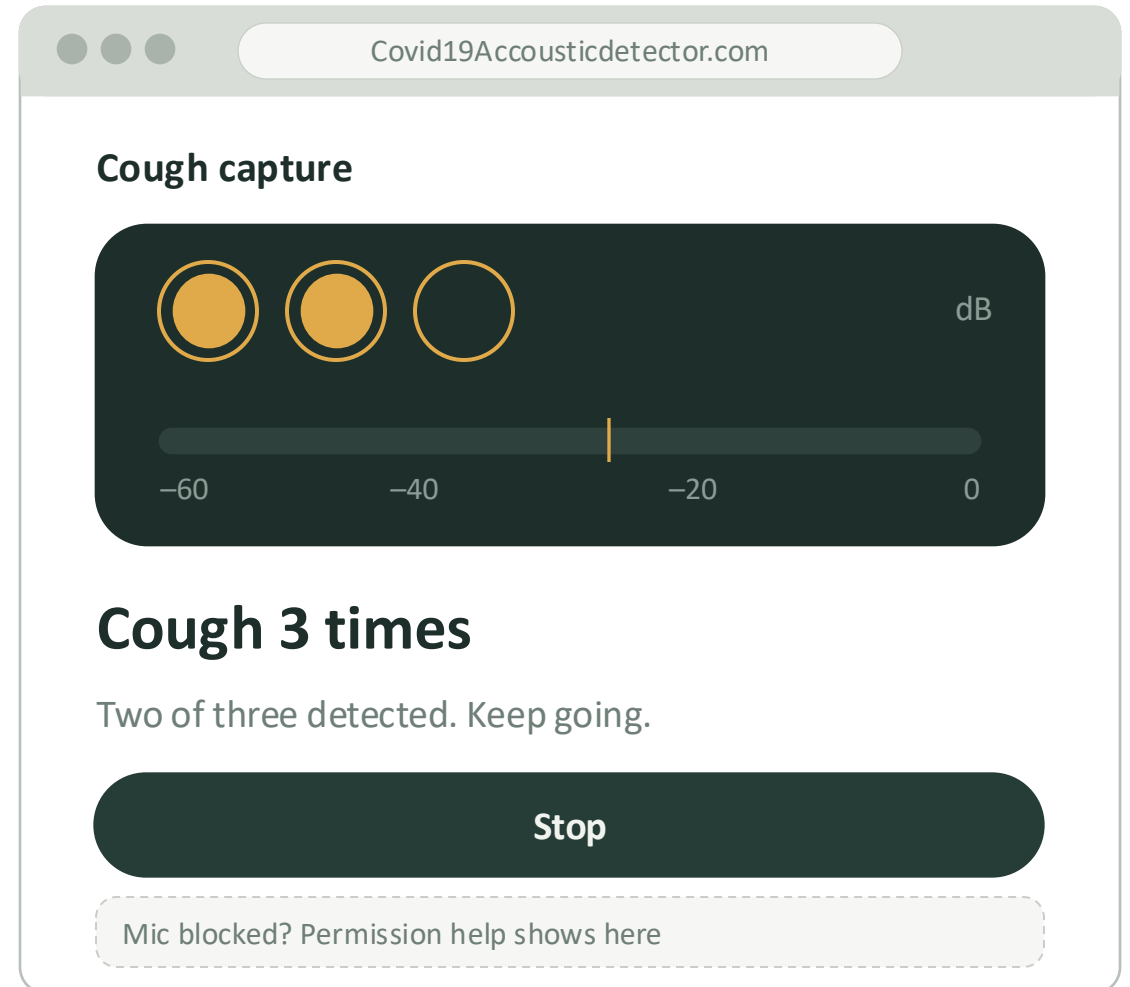
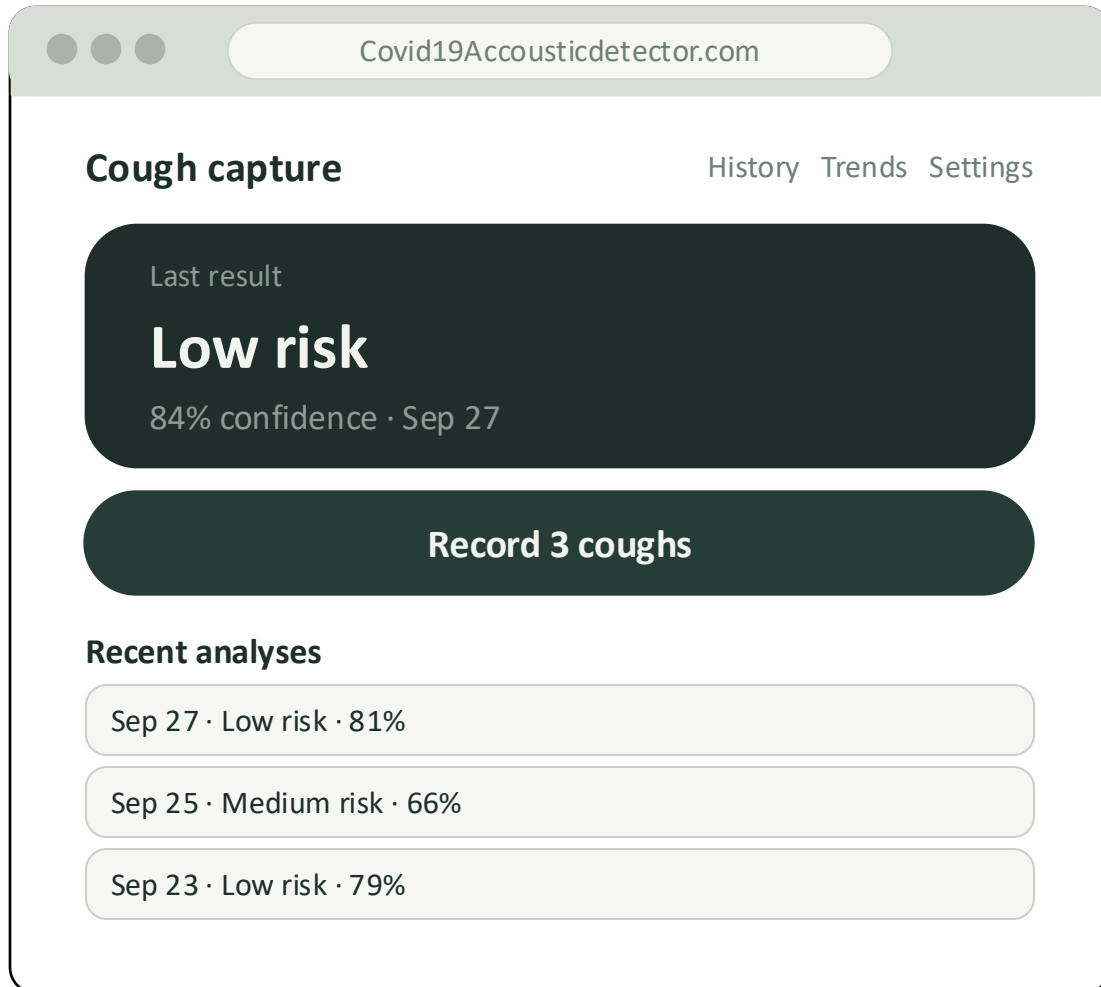


System Overview



Stateless server: no health data kept. History stays in the browser.

App Wireframe (1/2)



Built: recorder ends on "Sample captured." with Record another

App Wireframe (2/2)



Your result

MEDIUM risk
Confidence 62%

U

U

N

Votes: Uncertain 2, Negative 1

Consider testing, mask up around others, and monitor symptoms for 48 hours.

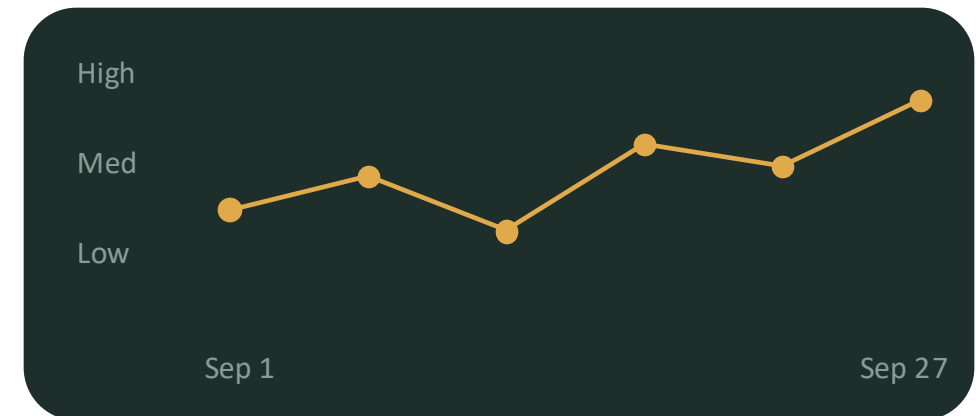
Cough detector: pass · SNR good

Pre-screening only. Not a diagnosis.

Record another



Trends



12 total

8 low risk

4 medium risk

Export CSV

Export JSON

Recorder Demo

- Guided three-cough recorder
- Live microphone level meter
- 16 kHz WAV via AudioWorklet
- Permission and error handling
- Analysis: coming soon

[link](#)



CNN Development

Goal

Identify patterns in
cough audio

3 Blocks

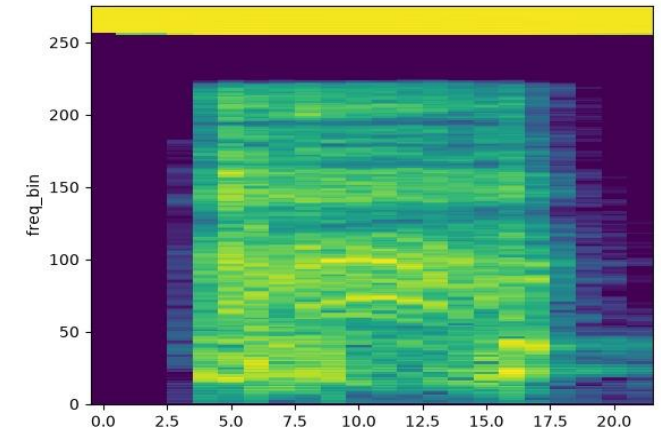
- Convolution -> BatchNorm -> ReLU
- Max Pooling reduces the data size

Output

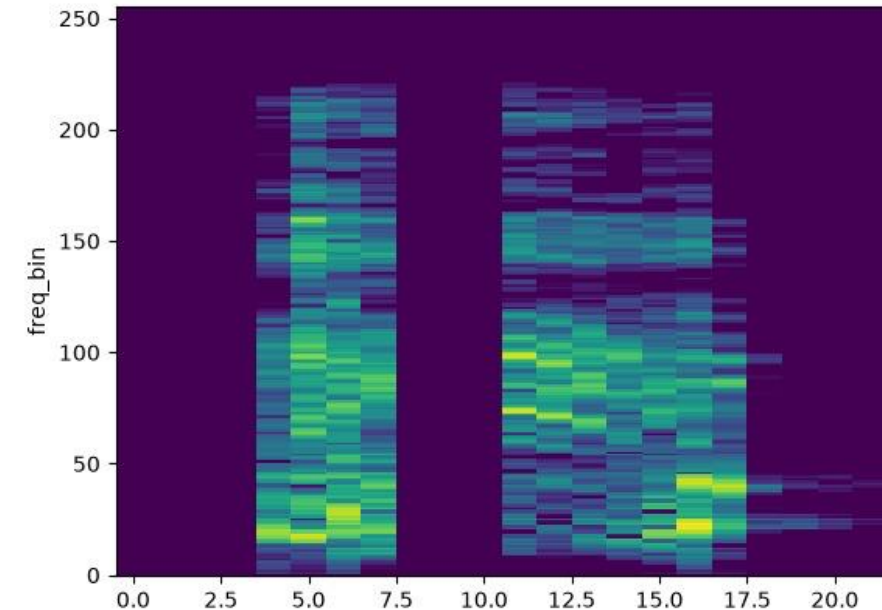
Healthy | Symptomatic | Covid-19

Current Stage – Baseline CNN Model Complete

Next: Training/Evaluation

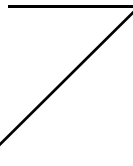


Class Imbalance



Milestone 2 Task Matrix (Oct 26)

Task	Drew	Amanda	Loleyi	Richard
Train & fine-tune CNN / ResNet transfer learning	●	●		
Model evaluation and reports (F1, ROC-AUC)	●		●	
FastAPI endpoints for upload and inference	●	●	●	
React UI for results and risk feedback				●



Questions?